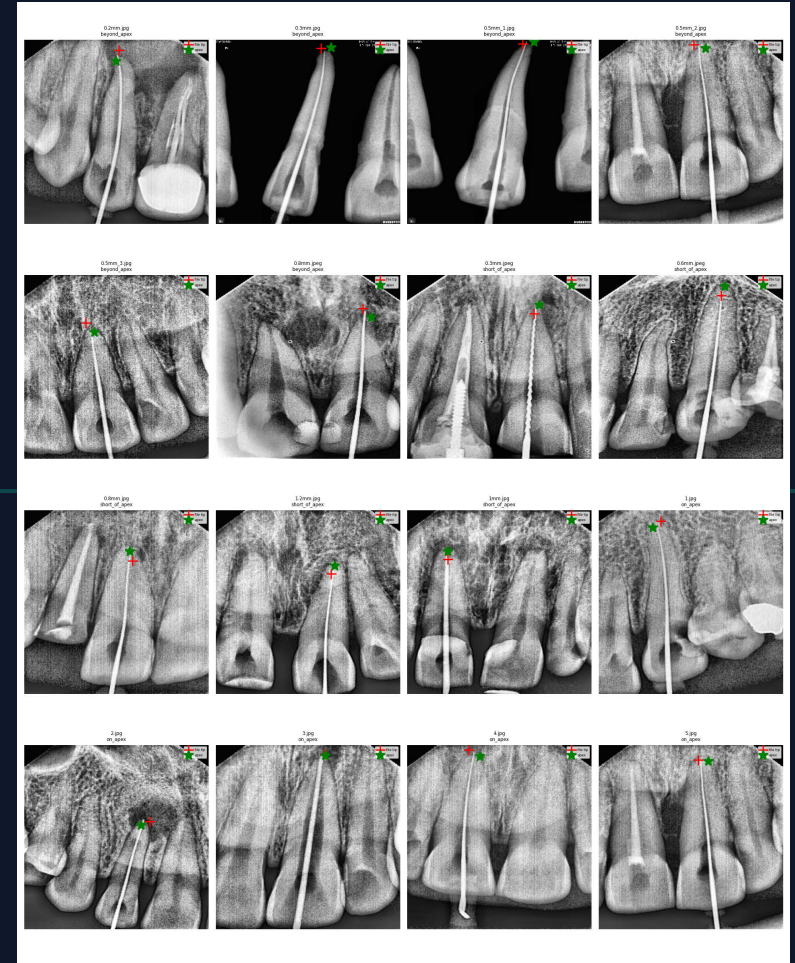


AUTOMATED ENDODONTIC FILE PLACEMENT

Classification and Offset Estimation
from Periapical Radiographs



PRECISION IN ROOT CANAL TREATMENT

-  Root canal success depends on precise file placement relative to the root apex.
-  Manual radiographic assessment is subjective and highly operator-dependent.
-  Automated pipelines provide objective classification and accurate offset estimation.

OBJECTIVE ASSESSMENT

Standardizing clinical workflows through AI-driven keypoint detection.

16 PERIAPICAL RADIOGRAPHS DATASET PROFILE

6

BEYOND APEX

5

ON APEX

5

SHORT OF APEX

Classification	Ground Truth Offset Range
Beyond Apex	0.2mm – 0.8mm
On Apex	0.0mm
Short of Apex	0.3mm – 1.2mm

AUTOMATED PIPELINE ARCHITECTURE

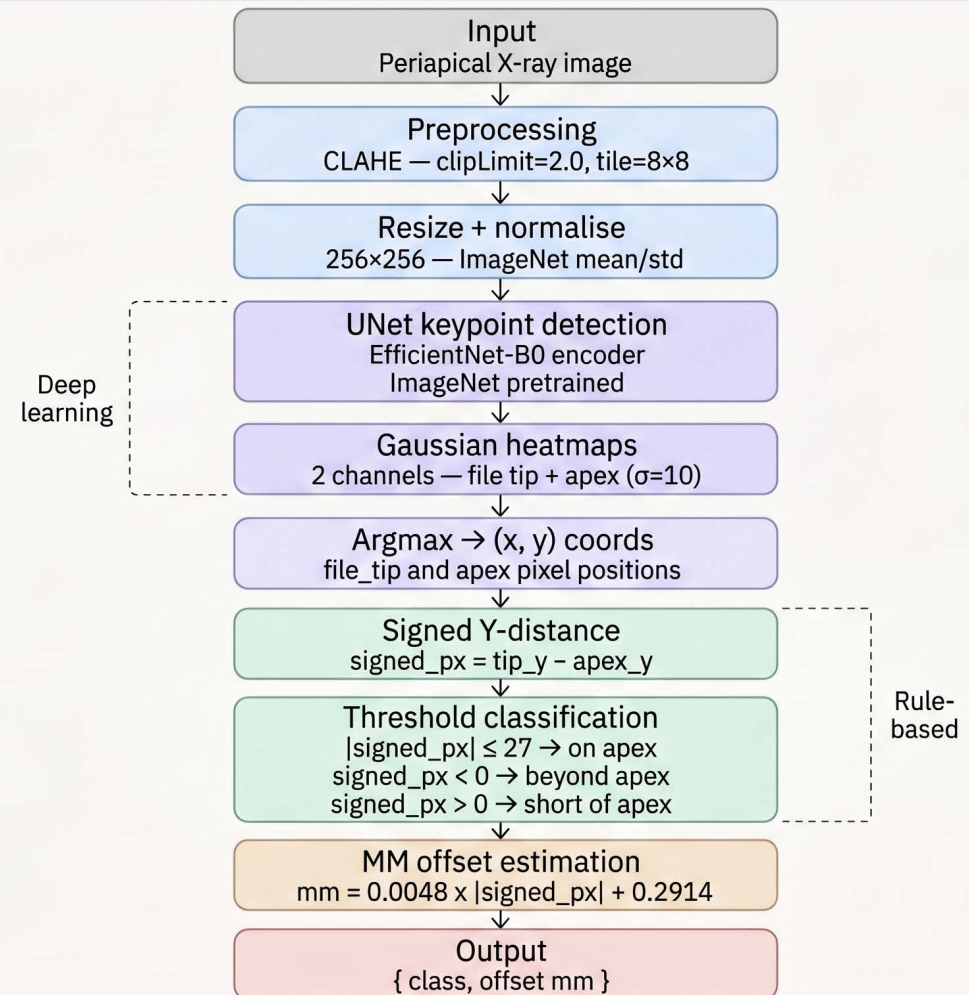
01 CLAHE Preprocessing & Image Resizing
(256×256)

02 UNet Keypoint Detection (EfficientNet-B0)

03 Gaussian Heatmap Regression (File Tip + Apex)

04 Signed Distance & Threshold Classification

05 Linear Regression for MM Offset Estimation



CONTRAST ENHANCEMENT WITH CLAHE

Raw periapical radiographs often exhibit low contrast, making it difficult to distinguish the metallic endodontic file from the surrounding root and bone structures.

ALGORITHM

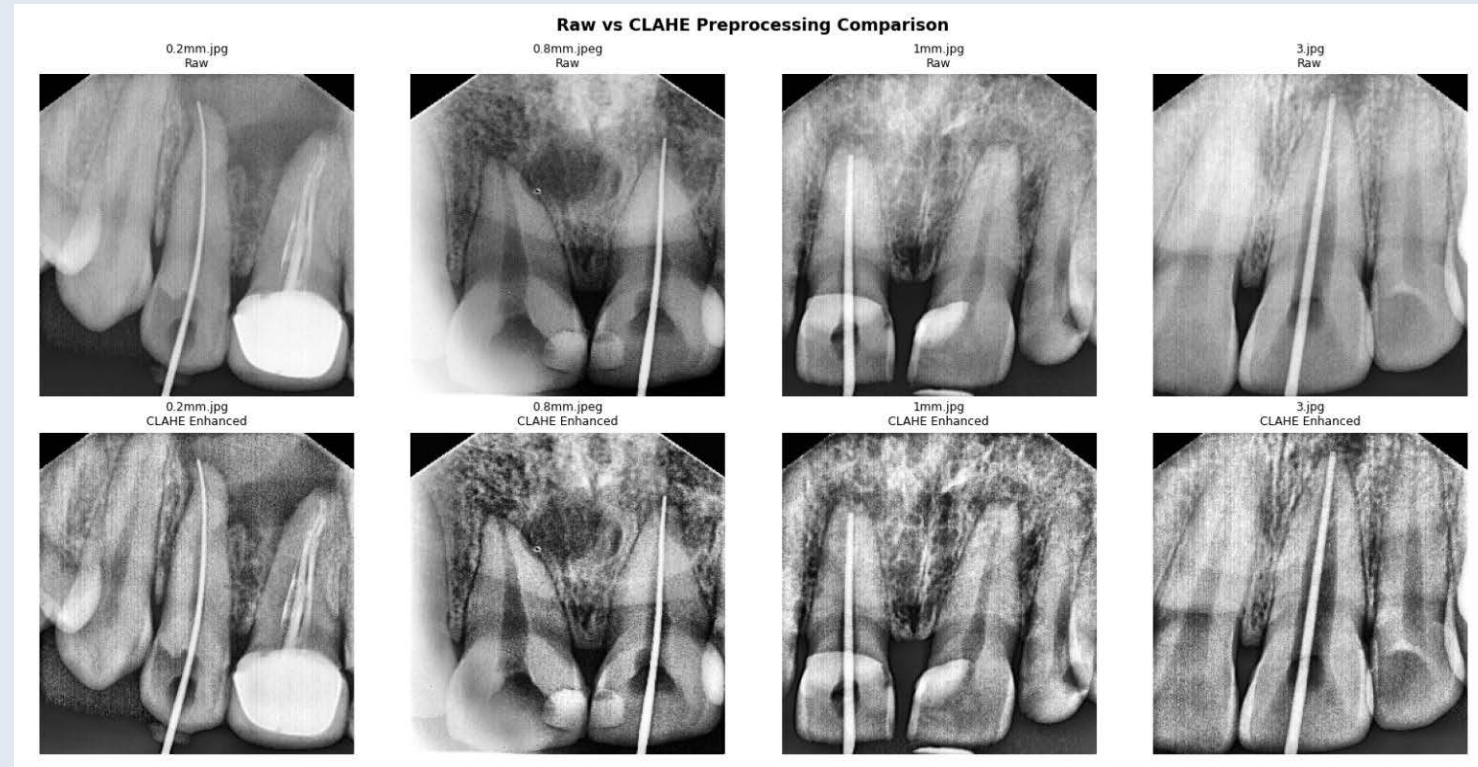
Contrast Limited
Adaptive
Histogram
Equalization

PARAMETERS

clipLimit: 2.0 |
tileGridSize: 8×8

RESOLUTION

Resized to 256 × 256 pixels



UNET WITH EFFICIENTNET-B0 ENCODER

ARCHITECTURE

UNet with EfficientNet-B0 encoder pretrained on ImageNet.

OUTPUT

2 Gaussian heatmaps ($\sigma=10$) for file tip and root apex localization.

TRAINING

500 epochs with a final training loss of ~ 0.0005 .

NOTE

Model has memorized the training set ($N=16$); generalization testing is pending.



FIGURE: PREDICTED KEYPOINT LOCALIZATION

CLASSIFICATION PERFORMANCE RESULTS

Class	Precision	Recall	F1-Score
Beyond Apex	1.00	0.67	0.80
On Apex	0.71	1.00	0.83
Short of Apex	1.00	1.00	1.00

Geometric Ceiling: 94% accuracy achievable with perfect keypoint localization.
Current model achieves 88% (14/16 correct) despite training set memorization.

88%
OVERALL ACCURACY

Classification Performance Metrics

Class	Precision	Recall	F1-Score
Beyond Apex	1.0	0.67	0.8
On Apex	0.71	1.0	0.83
Short of Apex	1.0	1.0	1.0

METRIC ANALYSIS: MM OFFSET ESTIMATION

LINEAR REGRESSION MODEL

ESTIMATION FORMULA

$$\text{mm} = 0.0048 \times |\text{signed_px}| + 0.2914$$

The model converts the signed pixel distance between the predicted file tip and the root apex into a clinical millimeter measurement.

PERFORMANCE METRICS

METRIC	VALUE
R ²	0.596
MAE	0.195mm
Tolerance	0.5mm

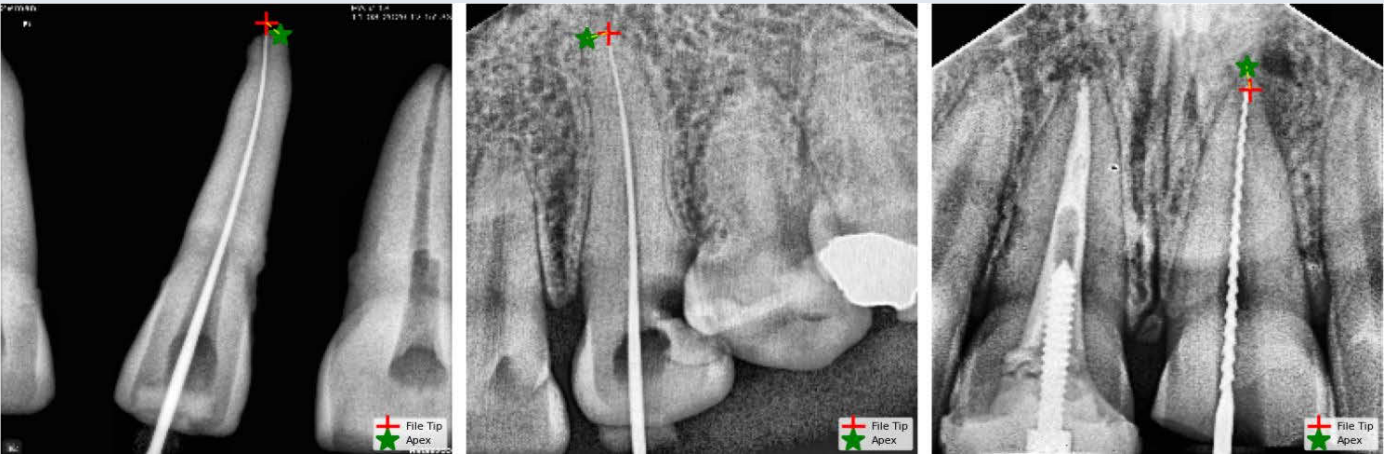
R² = 0.596 INDICATES MODERATE FIT

The Mean Absolute Error (MAE) of 0.195mm is significantly lower than the clinical tolerance, validating the pipeline's precision despite the moderate fit.

ERROR ANALYSIS & SUB-PIXEL LIMITATIONS

MISCLASSIFIED CASES (2/16)

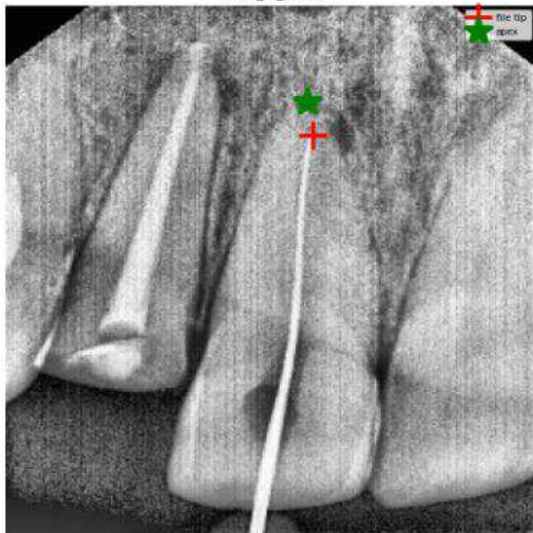
IMAGE	TRUE	PREDICTED	REASON
0.2mm.jpg	Beyond	On Apex	Offset is sub-pixel at 256×256 resolution; Y-coordinates were identical.
0.5mm_1.jpg	Beyond	On Apex	Model placed tip too close to apex (-18.3px), falling within the threshold.



FUTURE WORK

- Increase input resolution (e.g., 512 512) to resolve sub-millimeter offsets.
- Expand dataset size to improve model generalization beyond the training set.

0.8mm.jpg
short_of_apex



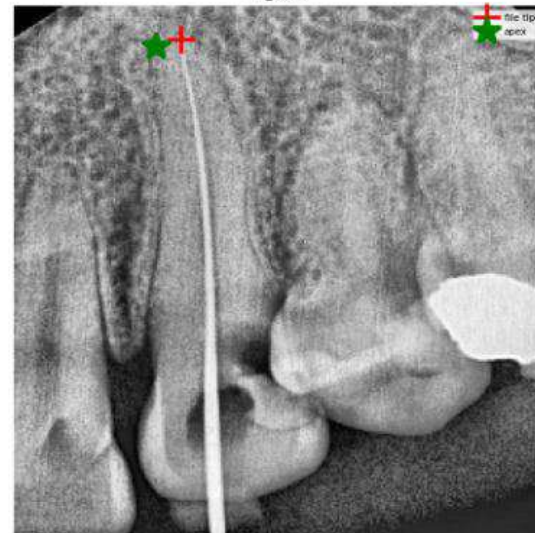
1.2mm.jpg
short_of_apex



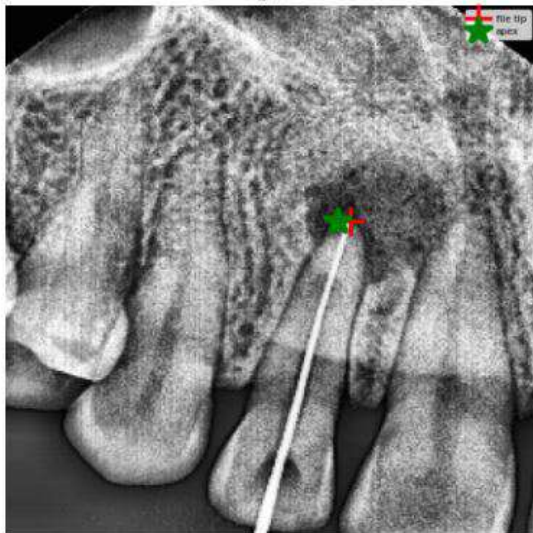
1mm.jpg
short_of_apex



1.jpg
on_apex



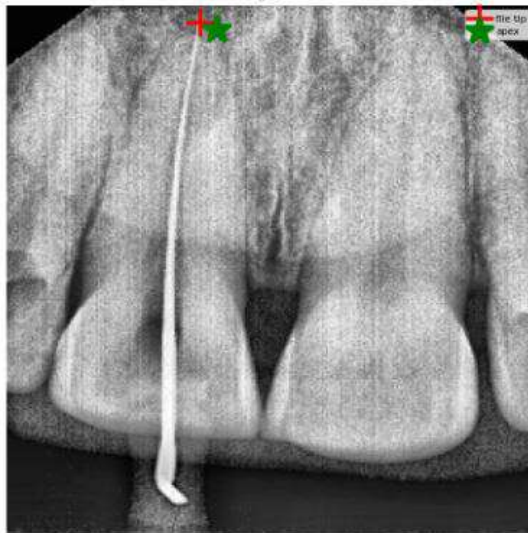
2.jpg
on_apex



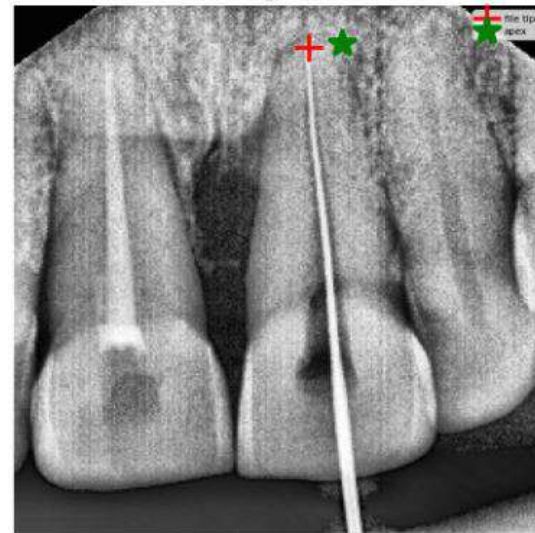
3.jpg
on_apex



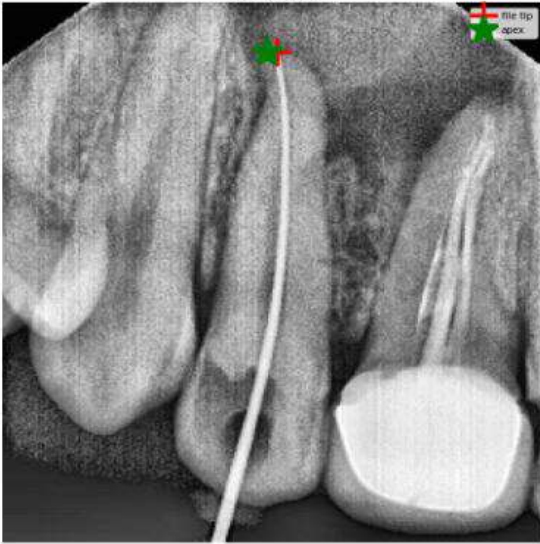
4.jpg
on_apex



5.jpg
on_apex



0.2mm.jpg
beyond_apex



0.3mm.jpg
beyond_apex



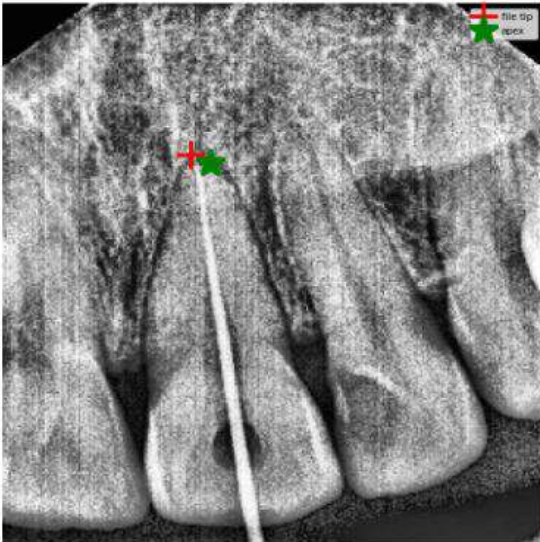
0.5mm_1.jpg
beyond_apex



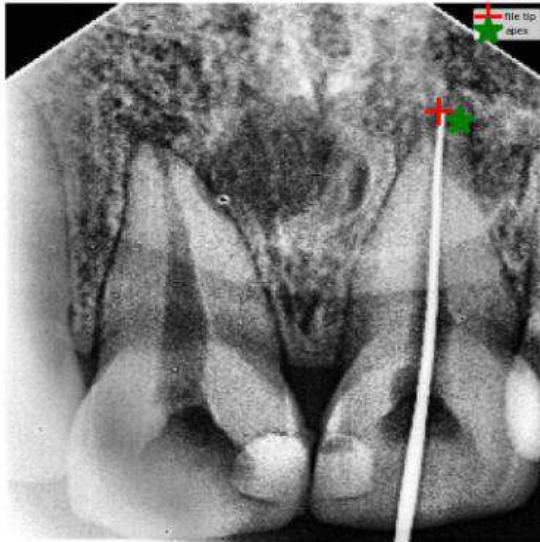
0.5mm_2.jpg
beyond_apex



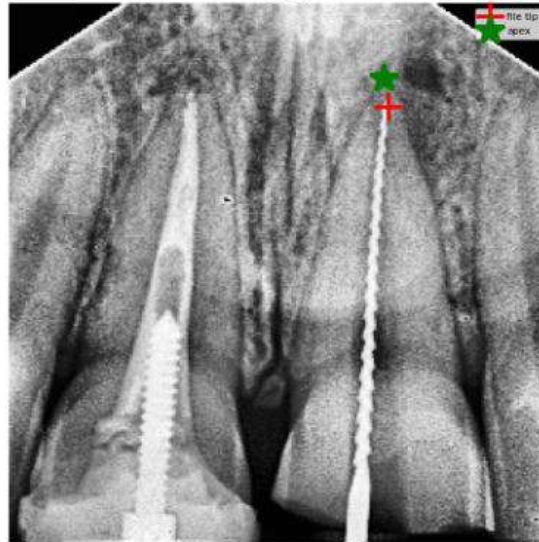
0.5mm_3.jpg
beyond_apex



0.8mm.jpg
beyond_apex



0.3mm.jpg
short_of_apex



0.6mm.jpg
short_of_apex

